

Solve in $\mathbb{R}$ the following equation:

$$\log_2 x + \log_2 (x+1) - 3 = 0$$

$$\log_2 (x(x+1)) = 3$$

$$\frac{\ln(x^2+x)}{\ln 2} = 3$$

$$x^2+x = 8$$

$$x^2+x-8=0$$

$$x_1 = 2.37 \quad \text{and} \quad x_2 = -3.37$$

$$x > 0 \Rightarrow -3.37 < 0 \text{ rejected}$$

$$\boxed{x = 2.37}$$

